

Deterministic Optimization

Large-Scale Optimization

Dantzig-Wolfe Decomposition

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Example

Large-Scale Optimization

Dantzig-Wolfe

Learning Objectives for this lesson

- Apply Dantzig-Wolfe decomposition to a numerical example

Example

- Consider the following LP:

$$\min \quad -4x_1 - x_2 - 6x_3$$

s.t.

$$3x_1 + 2x_2 + 4x_3 = 17$$

$$1 \leq x_1 \leq 2$$

$$1 \leq x_2 \leq 2$$

$$1 \leq x_3 \leq 2$$



Complicating constraint $Dx = b_0$



Easy constraint

- Easy constraint defines a polytope (a 3-dimensional box)

$$P = \{x \in \mathbb{R}^3 : 1 \leq x_i \leq 2, \forall i = 1, 2, 3\}$$

Let's denote the extreme points of P as $x^1, \dots, x^8$ (why 8 extreme pt?)

Set up Master Problem

- The master problem in the λ space:

$$\begin{aligned} (MP) \quad & \min_{\lambda_1, \dots, \lambda_8} \sum_{i=1}^8 \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\ & \text{s.t.} \quad \sum_{i=1}^8 \lambda_i (\mathbf{D} \mathbf{x}^i) = 17 \\ & \quad \sum_{i=1}^8 \lambda_i = 1 \\ & \quad \lambda_i \geq 0, \forall i = 1, \dots, 8. \end{aligned}$$

$$\begin{aligned} \mathbf{c} &= [-4, -1, -6]^T \\ \mathbf{D} &= [3, 2, 4]^T \end{aligned}$$

Start from a RMP

- Start with two extreme points $[1,1,2]^\top$ and $[2,2,2]^\top$
- We have

$$\mathbf{c}^\top \mathbf{x}^1 = [-4 \ -1 \ -6] \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = -22, \quad \mathbf{c}^\top \mathbf{x}^2 = [-4 \ -1 \ -6] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = -17$$

$$\mathbf{D}\mathbf{x}^1 = [3 \ 2 \ 4] \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = 18, \quad \mathbf{D}\mathbf{x}^2 = [3 \ 2 \ 4] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 13.$$

- (RMP) is given as

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \lambda_2} \quad -22\lambda_1 - 17\lambda_2 \\ & \text{s.t.} \quad 18\lambda_1 + 13\lambda_2 = 17 \\ & \quad \lambda_1 + \lambda_2 = 1 \\ & \quad \lambda_1, \lambda_2 \geq 0. \end{aligned}$$

Iteration 1: Solve RMP and Obtain Primal and Dual Solution

- Solve

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \lambda_2} \quad -22\lambda_1 - 17\lambda_2 \\ & \text{s.t.} \quad 18\lambda_1 + 13\lambda_2 = 17 \\ & \quad \lambda_1 + \lambda_2 = 1 \\ & \quad \lambda_1, \lambda_2 \geq 0. \end{aligned}$$

The basis matrix and its inverse are

$$\mathbf{B} = \begin{bmatrix} 18 & 13 \\ 1 & 1 \end{bmatrix} \quad \mathbf{B}^{-1} = \begin{bmatrix} 0.2 & -2.6 \\ -0.2 & 3.6 \end{bmatrix}$$

The optimal solution is $\begin{bmatrix} \hat{\lambda}_1 \\ \hat{\lambda}_2 \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} 17 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$. Form the dual variable:

$$[\hat{\mathbf{y}}^T \quad \hat{r}] = \mathbf{c}_B^T \mathbf{B}^{-1} = [-22 \quad -17] \mathbf{B}^{-1} = [-1 \quad -4].$$

Iteration 1: Check Optimality

- Solve the following Pricing Problem to compute min reduced cost:

$$\begin{aligned}\hat{Z} = \min \quad & \left(\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} \right) \mathbf{x} - \hat{r} \\ \text{s.t.} \quad & \mathbf{x} \in P\end{aligned}$$

Since $\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} = [-4, -1, -6] - (-1)[3, 2, 4] = [-1, 1, -2]$

$$\begin{aligned}\hat{Z} = \min \quad & -x_1 + x_2 - 2x_3 + 4 \\ \text{s.t.} \quad & 1 \leq x_1 \leq 2, 1 \leq x_2 \leq 2, 1 \leq x_3 \leq 2.\end{aligned}$$

The optimal solution is $\mathbf{x}^3 = [2, 1, 2]$, which is another extreme point of the cube P . $\hat{Z} = -1$.

This variable enters into the restricted problem. The associated column is $[\mathbf{D}\mathbf{x}^3, 1] = [16, 1]$, and the associated cost coefficient is $\mathbf{c}^\top \mathbf{x}^3 = -21$.

Iteration 2: Solve RMP

- Solve the new RMP:

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \lambda_2, \lambda_3} \quad -22\lambda_1 - 17\lambda_2 - 21\lambda_3 \\ & \text{s.t.} \quad 18\lambda_1 + 13\lambda_2 + 16\lambda_3 = 17 \\ & \quad \lambda_1 + \lambda_2 + \lambda_3 = 1 \\ & \quad \lambda_1, \lambda_2, \lambda_3 \geq 0. \end{aligned}$$

The optimal solution is $\hat{\lambda}_1 = 0.5, \hat{\lambda}_2 = 0, \hat{\lambda}_3 = 0.5$. The optimal basis matrix and its inverse are

$$\mathbf{B} = \begin{bmatrix} 18 & 16 \\ 1 & 1 \end{bmatrix} \quad \mathbf{B}^{-1} = \begin{bmatrix} 0.5 & -8 \\ -0.5 & 9 \end{bmatrix}$$

Iteration 2: Check Optimality

- Compute min reduced cost:

The dual variables are

$$[\hat{\mathbf{y}}, \hat{r}] = \mathbf{c}_B^T \mathbf{B}^{-1} = [-22, -21] \mathbf{B}^{-1} = [-0.5, -13].$$

$$\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} = [-4, -1, -6] - (-0.5)[3, 2, 4] = [-2.5, 0, -4]$$

The pricing problem is given as

$$\begin{aligned} \hat{Z} = \min \quad & -2.5x_1 - 4x_3 + 13 \\ \text{s.t.} \quad & 1 \leq x_1 \leq 2, \ 1 \leq x_2 \leq 2, \ 1 \leq x_3 \leq 2. \end{aligned}$$

Optimal $\hat{Z} = 0$, implying the current solution is optimal for the master problem!

Optimal Solution

- The final optimal solution:

$$\mathbf{x}^* = 0.5\mathbf{x}^1 + 0.5\mathbf{x}^3 = 0.5 \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + 0.5 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1.5 \\ 2 \end{bmatrix}$$

Summary

- We learned:
 - How to apply Dantzig-Wolfe decomposition to a real example